

The Influence of Varied Rest Interval Lengths on Depth Jump Performance

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ABSTRACT

The purpose of this study was to measure the effects of varied rest interval lengths on the vertical jump heights and ground reaction forces during the execution of a depth jump from a predetermined optimal height. The subjects were 12 men with a mean (SD) age of 25.08 ± 2.43 years. Each subject's optimal depth jump height was determined by executing depth jumps from 10–80 cm. After determining their optimal depth jump height, the subjects performed 3 sets of 10 depth jumps, each set with a different rest interval duration. The 3 rest intervals between depth jumps were 15, 30, and 60 seconds and were counterbalanced for each subject. Maximal vertical jump height and vertical ground reaction forces were calculated for each depth jump trial. The Peak Performance Motion Measurement System was used to measure vertical jump height and the Kistler force platform was used to measure ground reaction forces. Two-way analyses of variance revealed that rest interval length did not affect ($p > 0.05$) vertical jump height or vertical ground reaction forces. Therefore, this study demonstrated a 15-second rest interval was sufficient for recovery during the performance of depth jumps.

Key Words: stretch-shortening cycle, plyometrics, recovery, anaerobic

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Introduction

Plyometrics are exercises commonly used to enhance explosive power via the stretch-shortening cycle (SSC). The stretch component of the SSC refers to the eccentric muscle action, whereas the shortening component refers to the concentric muscle action (11). Elastic energy is stored in the tendomuscular system during the eccentric action (7). If the concentric action occurs immediately after the eccentric action, the muscles use the stored elastic energy (14) in combination with the normal voluntary tension to provide a more forceful contraction, which results in an increase in vertical jump height (11). To optimize the use of stored elastic energy, effective plyometric exercise should be

targeted at reducing the duration of the amortization phase (i.e., electromechanical delay between eccentric and concentric actions) and increasing the speed of eccentric action (within reasonably safe, controllable limits) (11). Studies examining different characteristics of the SSC have demonstrated increases in force and power production by decreasing the duration of the amortization phase and increasing the speed of eccentric action (3, 5, 6, 16, 22).

Countermovement plyometric jumps, which incorporate a deliberate SSC, enhance vertical jump height and force production compared with jumps that use concentric action only (1, 4, 15). Depth jumps optimize the use of the SSC by incorporating a greater load via dropping from a predetermined height. On landing from a depth jump, the muscles are stretched, storing mechanical energy in the tendomuscular system (7). The stored energy is now available for the subsequent shortening or concentric action (7).

Depth jumps are an exercise of short duration and high intensity; therefore, the primary energy system is the phosphagen system. Depth jumps have been demonstrated to be a more effective way to increase power output, force production, and jumping performance than other vertical jumping tests such as countermovement jumps (2, 11, 25). Determining an individual's optimal depth jump height is important (16, 25). An increase in height above optimal depth jumping height will decrease performance because of increased stretch load and eccentric work with corresponding decreased concentric work, resulting in a decreased ability to use stored elastic energy (2, 8). A decrease in height below optimal depth jumping height will decrease performance because of a decreased stiffness in the tendomuscular system and a decreased preactivation of the neuromuscular system (7). The phosphagen system supplies energy for high-intensity exercises that last up to approximately 10 seconds in the form of adenosine triphosphate (ATP) and creatine phosphate (CP) (17, 20). During this type of intense exercise and in the presence of the enzyme creatine kinase, phosphocreatine (PCr) and adenosine diphosphate interact to form creatine and ATP (13). Very intense lon-

Table 1. Physical characteristics of the subjects.

	Height (cm)	Weight (kg)	Leg length (cm)	Age (y)	Optimal depth jump (cm)
Mean ($\pm$ SD)	174.24 $\pm$ 6.28	82.13 $\pm$ 10.36	100.43 $\pm$ 6.40	25.08 $\pm$ 2.43	61.67 $\pm$ 16.42
Range	161.0–182.1	62.80–98.40	83.2–107.3	21–29	30–80

ger duration (i.e., 10–30 seconds) or repeated trials of very intense short duration exercises (i.e., 5–10 seconds) elicit a rapid decrease in the ability to generate power and force, possibly as a result of a reduced capacity to regenerate ATP due to a depletion of PCr and an accumulation of lactic acid (9, 19, 23). The depletion of PCr stores has been reported to occur during intense power exercises that require as few as 5 to 7 seconds per attempt (23). Thus, the rate of recovery between sets of shorter-duration explosive exercises is an essential short-term training variable that will influence performance and the resultant training adaptations (9).

Rest intervals of 3 to 4 minutes are generally recommended for recovery of the phosphagen energy system (90%) during high-intensity exercises, such as plyometric jumps and bounds (13, 18). For single-attempt depth jumps, however, a maximal repetition lasts less than 1 second and therefore may not completely deplete phosphagen stores. As a result, the regeneration of ATP via CP resynthesis may not be a primary factor of fatigue during depth jump training as is the case with other longer intense activities (19, 20). Partial replenishment of the phosphagen stores has been reported to occur rapidly, with as much as 70% restoration occurring in 30 seconds (12). Shorter rest intervals such as 15 to 30 seconds, therefore, may be sufficient during single-attempt depth jump training. The purpose of this study was to examine the effects of 3 different rest interval lengths on single-attempt depth jump performance from a predetermined optimal height.

Methods

Experimental Design

The subjects reported to the laboratory on 4 different occasions to do the following: (a) determine their optimal depth jump height, (b) perform 10 depth jumps from the optimal height with a 15-second rest period, (c) perform 10 depth jumps from the optimal height with a 30-second rest period, and (d) perform 10 depth jumps from the optimal height with a 60-second rest period. Each visit was separated by 24 hours of rest (i.e., no exercise). The order of the rest interval lengths used in testing was varied systematically among subjects to eliminate an order effect. Before performing the depth jumps, subjects performed a general warm-

up and received 3 practice depth jumps off a 20-cm bench or box. Vertical jump heights and ground reaction forces were measured during the 10 depth jump trials.

Subjects were instructed not to exercise for the 24 hours preceding all testing sessions.

Subjects

Twelve male volunteer university students with a mean (SD) age of 25.08 $\pm$ 2.43 years were the subjects used in this study (Table 1). The subjects were recreational weightlifters with a minimum of 1 year of weightlifting experience as determined by a health screening questionnaire. None of the subjects were competitive lifters or athletes; however, all of them could squat 1½ times their body weight and leg press 2 times their body weight. The subjects signed a human subjects informed consent form before participating in this study. All of the testing was performed in the Exercise Physiology/Biomechanics Laboratory at San Jose State University.

Optimal Depth Jump Height

To determine optimal depth jump height, each subject executed a series of depth jumps from different heights. The heights ranged from 10–80 cm, increasing by 10 cm for each jump. The subjects received a 3 minute rest period between each depth jump. The height in which they achieved their highest rise in their iliac crest was their optimal depth jump height (16). The range of the subjects optimal depth jump height was 30–80 cm, with a mean height of 61.67 cm.

Vertical Jump Height

The Peak Performance Motion Measurement System was used to measure the vertical jump height. Included within the Peak Performance Motion Measurement System were the following: HIQ personal computer, peak 5.0 software, Panasonic AG 6300 VCR, Panasonic CT 1400 MG color video monitor, and the Panasonic AG 450 video camera, which tapes at 60 fields per second. The video camera was 1.3 m high and 5 m from the force platform. It was positioned perpendicular to the sagittal plane. The iliac crest of each subject was marked with tape and digitized, and through video analysis the height rise of the iliac crest was determined. The difference between each subject's iliac crest while standing and the height of their iliac crest at peak flight was determined as their vertical jump

Table 2. Vertical jump heights from optimal depth jump at each rest interval.†

Jump	Rest interval length (s)		
	15	30	60
1	40.9 ± 6.2	41.2 ± 6.0	39.5 ± 6.3
2	40.2 ± 6.7	39.7 ± 6.4	39.5 ± 4.9
3	40.2 ± 5.3	39.2 ± 4.9*	39.6 ± 5.2
4	40.1 ± 5.9	39.2 ± 5.6*	40.3 ± 4.7
5	40.7 ± 4.4	41.1 ± 4.9	40.1 ± 5.1
6	40.0 ± 5.8	39.7 ± 6.3	40.6 ± 4.3
7	41.4 ± 4.9	41.2 ± 6.4	40.5 ± 4.8
8	41.0 ± 5.5	40.8 ± 5.9	38.6 ± 5.2
9	41.0 ± 5.8	40.4 ± 5.9	39.3 ± 5.6
10	42.5 ± 5.0	41.5 ± 5.7	39.6 ± 5.5

† Values are mean ($\pm SD$) reported in centimeters.

* $p < 0.05$ vs. corresponding jump trial number 10.

height. The rise of the iliac crest was measured in centimeters.

Ground Reaction Forces

The mean vertical ground reaction forces were recorded for each of the 10 jump trials for the 15, 30, and 60-second rest interval lengths. A Kistler force platform, model 5004 dual model amplifier, was used to measure the ground reaction forces during the depth jump. The amplifier was set at 1–11, which collected 600 samples of force per second for 3 seconds before and after jumping. The Kistler force platform was manually triggered and interfaced to the Peak Performance Motion Measurement System. The ground reaction forces were measured in Newtons in the Z direction during take-off. Some of the forces were not calculated by the Kistler force platform; therefore, only 6 complete sets of data were recorded for ground reaction forces.

Rest Interval Lengths

The rest interval lengths were counterbalanced for each subject, with a total of 6 combinations for the rest interval order, with 2 subjects in each group. The rest interval orders were as follows: 15, 30, 60; 15, 60, 30; 30, 15, 60; 30, 60, 15; 60, 30, 15; and 60, 15, 30 seconds. All subjects were tested using each rest interval length: 15, 30, and 60 seconds.

Statistical Analyses

Descriptive statistics were calculated and reported for the vertical jump heights, ground reaction forces, optimal jump height, age, height, body weight, and leg length. Two-way analyses of variance were used to identify potential differences in the vertical jump heights or ground reaction forces between the 3 different rest interval lengths. One-way analyses of variance with Tukey post hoc tests were used for pairwise comparisons when a significant F was found. The α level was set at 0.05 for this study.

Results

Vertical Jump Heights

No differences were observed in jump heights among 15, 30, and 60-second rest trials for any of the jumps (1–10) (Table 2). Within the group of jumps using 30-second rests, jump heights during trials 3 and 4 were significantly less than the respective jump height for trial 10. Jump heights did not differ among jumps 1–10 within the 15 or 60-second rest groups.

Vertical Ground Reaction Forces

No differences were observed in ground reaction forces among 15, 30, and 60-second rest trials for any of the jumps (1–10) (Table 3). Also, ground reaction forces did not differ among jumps 1–10 within groups (15, 30, or 60 seconds).

Table 3. Vertical ground reaction forces from optimal depth jump at each rest interval.*

Jump	Rest interval length (s)		
	15	30	60
1	7745.04 ± 1021.64	7178.59 ± 1012.50	7194.77 ± 974.75
2	7784.66 ± 844.74	7920.65 ± 1135.23	7699.00 ± 1274.24
3	7616.55 ± 792.25	7964.55 ± 763.77	7694.71 ± 1205.92
4	7549.09 ± 738.88	7875.68 ± 1222.76	7161.50 ± 1483.60
5	7687.22 ± 774.63	7741.83 ± 716.94	7998.82 ± 1284.29
6	7112.13 ± 811.85	7639.03 ± 533.39	7703.28 ± 1505.88
7	7754.68 ± 927.39	7888.48 ± 325.40	7861.76 ± 1195.87
8	7544.33 ± 845.61	7890.67 ± 848.00	7866.04 ± 1227.86
9	7639.03 ± 1319.64	7977.40 ± 1449.63	7605.84 ± 1113.37
10	7605.84 ± 1116.63	7933.50 ± 1626.28	7370.26 ± 1417.46

* Values are mean ($\pm SD$) reported in newtons.

Discussion

The primary finding of this study was that rest interval length during single-attempt depth jumps did not influence vertical jump height or ground reaction force. The duration of an exercise directly influences muscular fatigue (9). A single, maximal depth jump lasts less than 1 second and therefore does not completely deplete the phosphagen stores. Harris et al. (10) demonstrated that maximal exercise tests to exhaustion lasting 2–6 minutes caused an 84% decrease in CP concentration following dynamic exercise and an 85% decrease following isometric exercise. Even when the phosphagen stores are completely depleted, it can take 3–5 minutes to replenish ATP, with complete CP restoration occurring after 8 minutes (18). Since a depth jump does not deplete the phosphagen stores, full recovery during the present study was not required for the regeneration of ATP or CP.

The intensity of exercise is also a determining factor in fatigue (9). Sahlin and Ren (19) demonstrated that maximal isometric force 15 seconds after fatiguing contractions was 80% of the maximal force before the contractions. Strength levels following isotonic contractions were 84% of initial strength values after 35 seconds of rest (21). A 60-second rest interval length was used following maximum attempts in the bench press, and the results demonstrated that there was no significant difference among trials (24). Previous research indicates that rest intervals of 1 minute and even 15 seconds allow for recovery from maximal exercise (19, 24). Our data support the results of the aforementioned studies, since the 3 rest interval lengths did not significantly affect force production and vertical jump capabilities. Even though Sahlin and Ren (19) and Stull and Clarke (21) performed fatiguing exercises before the strength tests, their results still demonstrated a recovery of strength following the same rest interval lengths used in the present study. The experimental design of Weir et al. (24) was similar to the present study since the 1 repetition maximum in the bench press was not preceded by fatiguing contractions, and neither the present study nor the study by Weir et al. (24) reported significant differences following a 60-second rest interval length. Further, the present study has demonstrated that 15 seconds is a sufficient amount of time for complete recovery during depth jumps at maximal effort, since vertical jump height did not decrease across the 10 depth jump trials. Fatigue did not appear to be present in the recreational weightlifters performing single-attempt depth jumps. Therefore, shorter rest intervals with the range of 15–60 seconds during single-attempt depth jump training may shorten overall training time while maintaining the quality of the exercise.

During the 30-second rest interval group, a significant difference was observed between jumps 3 and 10

and 4 and 10 (Table 2). Both jumps 3 and 4 were significantly lower in height than jump 10, suggesting that fatigue had not occurred by jump 10. These findings probably represent an anomaly in the data collection since the third and fourth jumps using a 30-second rest interval did not differ from the third and fourth jumps using either a 15- or 60-second rest interval. Further, there were no other significant differences between jumps 3 and 4 and any of the other jumps during the 30-second rest interval.

Since no differences occurred in this study in vertical jump heights or ground reaction forces using rest intervals ranging from 15–60 seconds, neuromuscular functioning did not appear to be either physiologically or biomechanically impaired. Hence, it appears that high-intensity plyometric training using exercises such as single-attempt depth jumps can be performed with rest intervals as short as 15 seconds without effecting neuromuscular performance. These results suggest that optimal depth jump training can be performed with a short rest interval, thereby decreasing the work-rest ratio in a training session and optimizing sports-specific conditioning.

Practical Applications

Any coach or professional who uses plyometrics in their training program should be aware of the physiological effects of plyometrics to maximize both performance and safety. One way of maximizing performance is to ensure the athlete has sufficient rest between sets to allow the regeneration of the phosphagen stores. It was demonstrated in this study using recreational weightlifters that 15 seconds of rest was a sufficient recovery period for consistent performance in the depth jump, as measured by the vertical jump heights and vertical ground reaction forces. Hence, other plyometric exercises of similar intensity and duration can also use 15-second rest periods. Plyometric exercises of longer duration, such as multiple jumps and bounding, may need to use longer rest periods. Depending on the number of jumps, rest periods of 30–60 seconds may need to be used. Further, the results of this study should be validated with different types of athletes, and thus the generalization of these results should be limited to the experimental design of this study.

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